CGRA259 - Caylum Blenkhorn - 300501291

GAME REPORT 2 (Group & Individual)

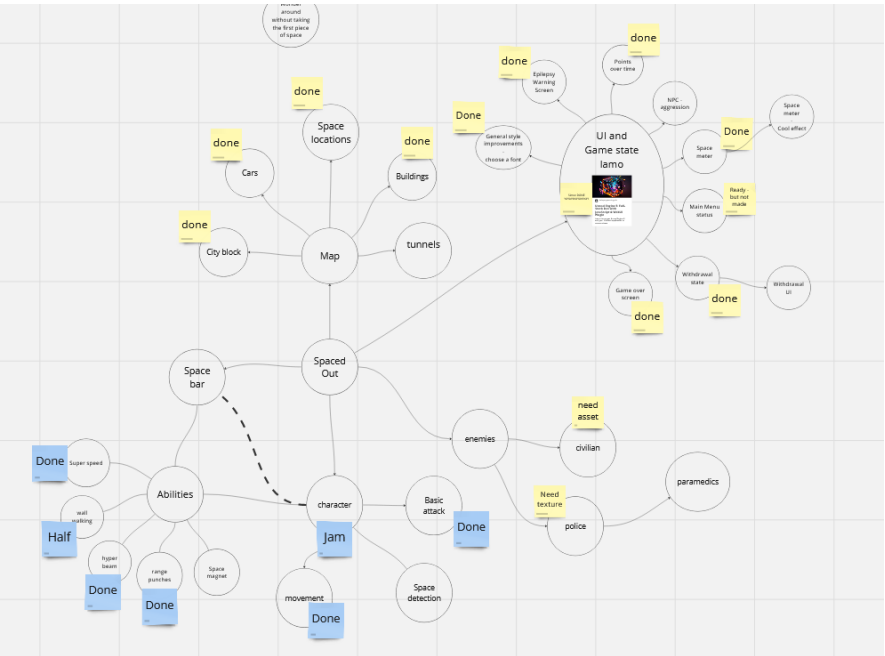
~~ Part 1: Group ~~

To preface, our group is comprised of myself (Caylum) and a few of my friends (Simon and Jamie), who are not enrolled in CGRA259 course.

The second Jam our group participated in was the 54th **Ludum Dare**, running from the 29th of September to the 2nd of October. The theme was '**Limited Space**', with a timeframe of 72 hours. Being a full day longer than the first Jam, and now with some experience participating in jams, our group decided to have an MVP ready to go 24 hours before the deadline. At the start of the jam, 1-2 hours was designated to developing our concept. Even with the hours set aside, our group came up with a concept almost immediately.

Like the first Jam, we believe the timeline we set was followed well, having a concept ready to go that we all felt passionate about drastically helped speed up the early stages of the Jam. Towards the end, we had an MVP done approximately 24 hours before the deadline, and with that time we could spend the remaining day bug fixing and adding features we felt necessary.

Instead of the main idea we believed groups would go for with the theme 'Limited Space', in the sense space is limited in terms of restricting movement. Our team took on the idea of 'space' being the name of a resource of some kind that the player has limited supply of.



Our group decided to use Unreal Engine 5.2, the most recent version at the time. Like the previous Jam, the members of the group had moderate experience using it, for version control, we decided to use GitLab over AzureDev, as we had more experience using Git. Most of us also used Github Desktop. Additionally, we used Miro.com to create a brainstorm of how our game would function, and what pieces were required to make it work. This also allowed us to manage who was working on what, and what had been completed simply by a glance at the board.

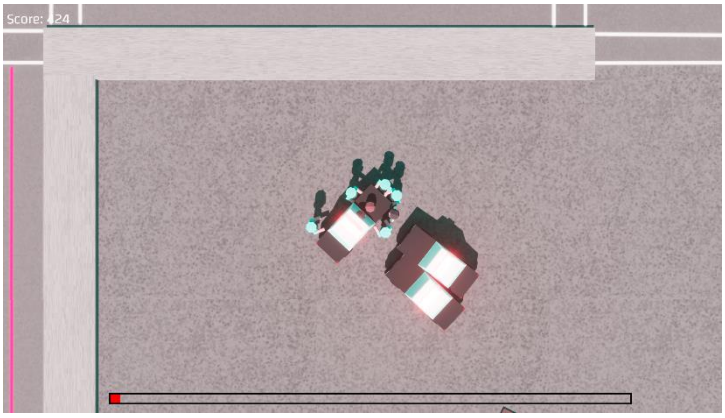
Other important tools used included Javascript for the UI, Blender and 3DCoat for the art and assets.

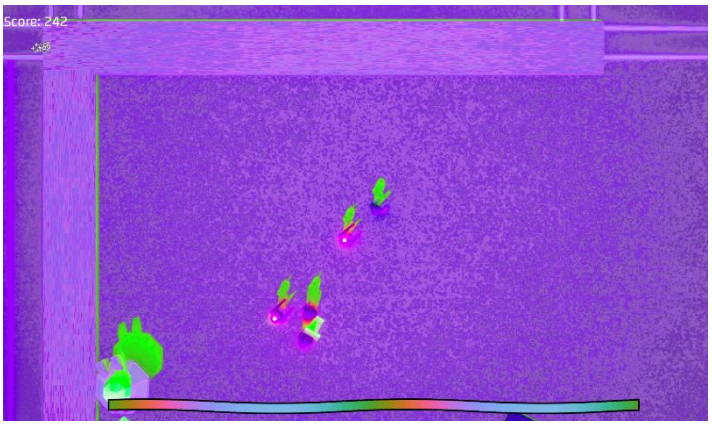
Akin to the previous Jam, none of us considered adding audio to our game, we decided to focus our attention on developing the rest of the game and not bother with audio, as none of us had much experience in audio design. Given more time, the group felt adding audio would greatly enhance the user experience.

For assets, since we chose 3d over 2d, we decided to use simple low poly models with quick low-res textures applied to them. Like pixel art, this allowed us to very quickly fabricate new models and textures for the game.

The gameplay loop of our concept and game was relatively simple, keep collecting 'space', a psychedelic item that respawns around the enclosed game world town, and keep your 'high' going for as long as possible. The player is required to run around the town and enter buildings while avoiding civilians and police pushing the player around and away from 'space'.

The higher the player's high meter is, the more trippy and visually striking the game becomes, the UI warps around, and the contrast and colours begin to shift and warp, when sufficiently high, the police and civilians change into aliens, and the player gains abilities like firing lasers and beams, pushing enemies away.





The game uses a very basic AI in it's npcs to pathfind towards the player, using the AIMove to Node, we can input the player's location and utilizing a NavMesh over the entire map, pathfind towards the player. The police cars do the same, however when getting close enough to the player and stopping two police officers pop out from the car. An issue we encountered when doing this is the officers would try to pathfind through the police car and get jammed. We were required to make the NavMesh dynamic, so it could be updated during runtime, and add a NavModifier component to update the NavMesh and make a radius around the car invalid locations for the AI to move to.

The main unique system of note in our game is the UI, instead of using Unreal Engine's widget-based system, it uses a webpage as the UI, and places it overtop the game. The decision to do this was the choice of Simon, who is a full time Stack Developer and is proficient in web design. The aim was to make the UI of the game visually striking and unique, with effects such as wobbling the high bar, more than we believed we were capable of doing in unreal's widget system.

This is implemented through a custom blueprint to Web pipeline written half and half in blueprints and javascript. Anytime we want to send and receive information to and from the web interface, we wrap strings in JSON objects, which the web listens to. The web UI itself is something we can reference as an object in the blueprint.

In testing we found we were fully capable of using the game as a browser, being able to google search things, watch youtube, ect. There are many applications for this web interface and we plan to explore this in future Jams and projects

Our group allocated roles:

Caylum: World building, Enemies, Visuals

Simon: Web, UI, Supporting

Jamie: Art, Player

Like the previous Jam, roles were allocated simply by what each group member wanted to work on most, ensuring each member was passionate about what they were working on. We felt this was a good decision to continue with, as it worked well prior.

~~ Part 2: Individual ~~

1) Teamwork

communication in the group was very effective and consistent, we were in a discord call for the majority of the jam, and frequently conversed to each other about where we were at, what needs attention, and give help if needed.

Tasks were allocated through us all asking what each member wanted to, a member who is forced to work on something they aren't interested in I believe is detrimental to

fast success, so each member chose something they had interest or passion in. In addition to having tight communication, help was readily available, and tasks could be swapped around if a member found themselves in a roadblock or got stuck.

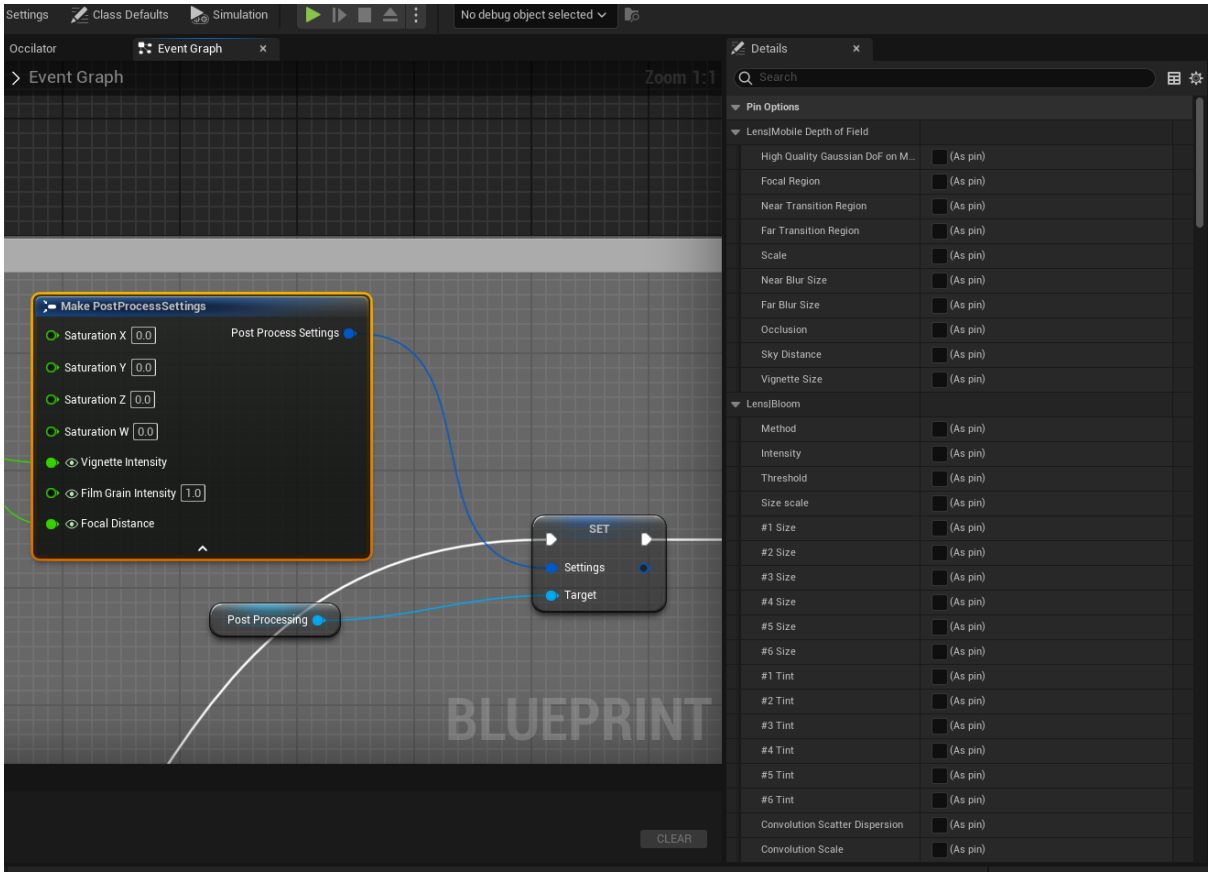
Team member strengths also played a role in task allocation, each member working on tasks they were familiar in or with was very helpful.

Like the first jam, I personally would not change how our team undertook task allocation; I feel it was very successful, and plan to use the same for the final jam.

2) Individual

Detailed above, my contribution to the project was through npcs and their AI navigation, I also made the car movement and logic to ensure they don't crash into each other when driving up and down the streets. I also worked on the visual effects that occur when the player gets high, this is done through the modifying the post processing the camera is using. Depending on the player's state, the visuals wave in and out with varying intensities changing via an oscillating wave function.

Learning how the Navigation Meshes work was very useful to learn, initially I chose a more complex system, being an AI behaviour tree. While this was infinitely more customizable and potentially the better option for expanding the game in the future, I felt it was faster and more efficient to run the basic AIMoveTo node. The police car npc uses a simple behaviour tree switching between pursuit and stopping.



Post processing has many different customizable features.

3) Changes

I believe our second Game Jam as a group went very well, again we followed the timeline we set ourselves, having the MVP ready early gives us lots of opportunities to bugfix, add new features and polish, which we did having had the time.

I found we tended to struggle with UI navigation, our previous and current game jam had no features I believe necessary for a game this being: Options, Controls, Some form of information / help section, and additionally credits of some kind.

In the final jam, we plan to add audio for the first time, as we believe it a vital missing component to our Games.